

FCU3

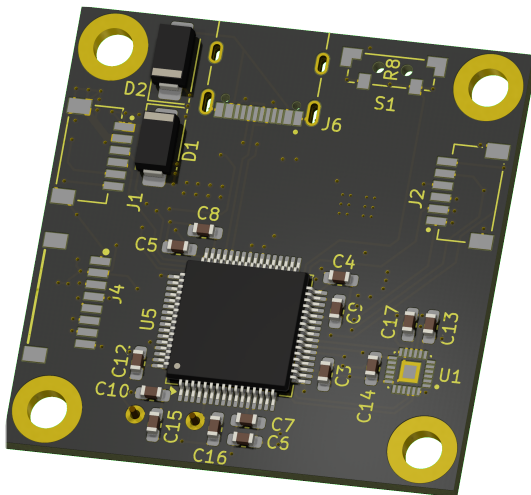
Variant: PRELIMINARY

2025-11-30

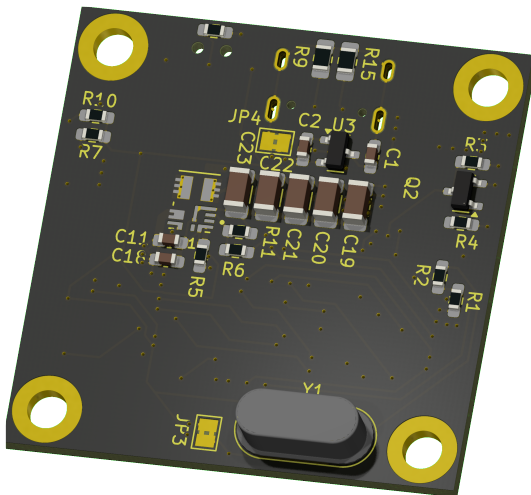
Rev + (Unreleased)

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TOP VIEW



BOTTOM VIEW



NOTES

Comment

Not fitted components are marked as **X**

DRAFT - Very early stage of schematic, ignore details.
PRELIMINARY - Close to final schematic.
CHECKED - There shouldn't be any mistakes. Contact the engineer if you find any.
RELEASED - A board with this schematic has been sent to production.

Date: 30-Nov-2025

DESIGN CONSIDERATIONS

DESIGN NOTE:

Example text for informational design notes.

DESIGN NOTE:

Example text for debug notes.

DESIGN NOTE:

Example text for cautionary design notes.

DESIGN NOTE:

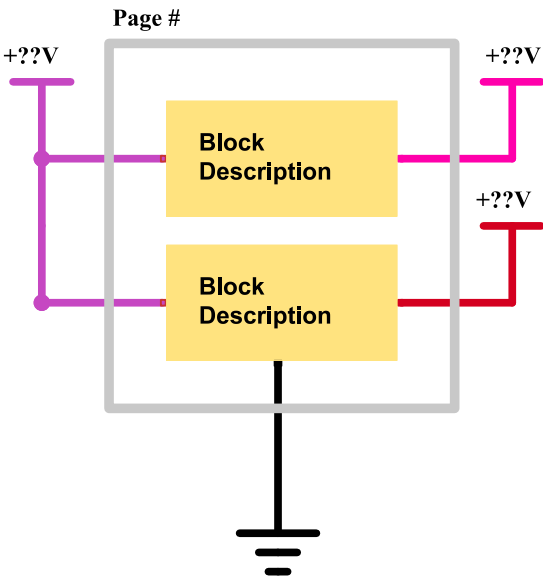
Example text for critical design notes.

LAYOUT NOTE:

Example text for critical layout guidelines.

	Comments:	Company: Kaldstrand Electronics	Variant: PRELIMINARY	Git Hash: 4dad50
	Sheet Title:	Board Name: FCU3	Project Name: Flight Controller 3	
	Sheet Path: /	File Name: FCU3.kicad_sch	Designer: Kaldstrand	Date: 2025-01-12
			Revision: + (Unreleased)	
			Size: A3	Sheet: 1 of 8

[2] Block Diagram

Page #
+??V


+??V
Page # SECTION
BLOCK A Port Name ???
Page # BLOCK B PORT C Port Name ???
+A3V3 BLOCK D Port Name ???
Page # BLOCK E Port Name ???
BLOCK F Port Name ???

SECTION
???
???
Port Name
Port Name
Port Name
Port Name
PORT NAME

Page # SECTION
BLOCK G
BLOCK H
BLOCK I
BLOCK J

Target specifications:

Input voltage:	?? - ?? V
Spec 2	??
Spec 3	??
Spec 4	??

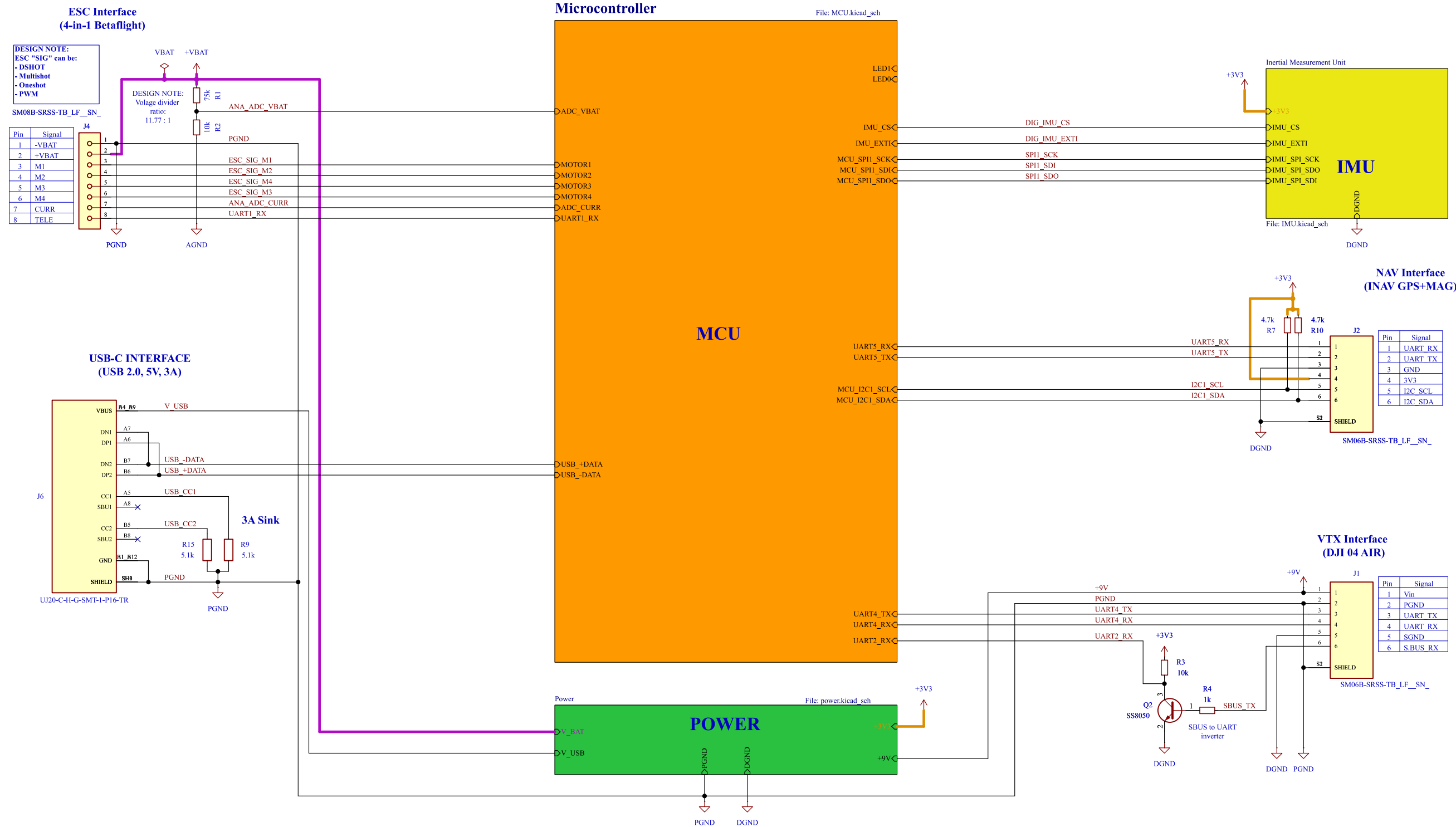
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		Board Name: FCU3	Project Name: Flight Controller 3	
	Sheet Title: Block Diagram	File Name: Block Diagram.kicad_sch	Designer: Kaldstrand	Date: 2025-01-12
	Sheet Path: /Block Diagram/	Reviewer:	Size: A3	Sheet: 2 of 8

Input voltage: ?? - ?? V

Spec 3 ??

	Comments:	Company: Kaldstrand Electronics		Variant: PRELIMINARY		Git Hash: 4dadb50	
		Board Name: FCU3		Project Name: Flight Controller 3			
	Sheet Title: Block Diagram	File Name: Block Diagram.kicad_sch	Designer: Kaldstrand	Date: 2025-01-12	Revision: + (Unreleased)		
	Sheet Path: /Block Diagram/		Reviewer:	Size: A3	Sheet: 2 of 8		

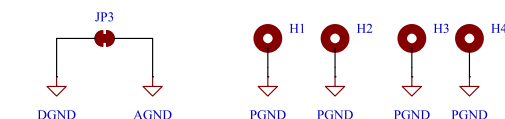
[3] Project Architecture



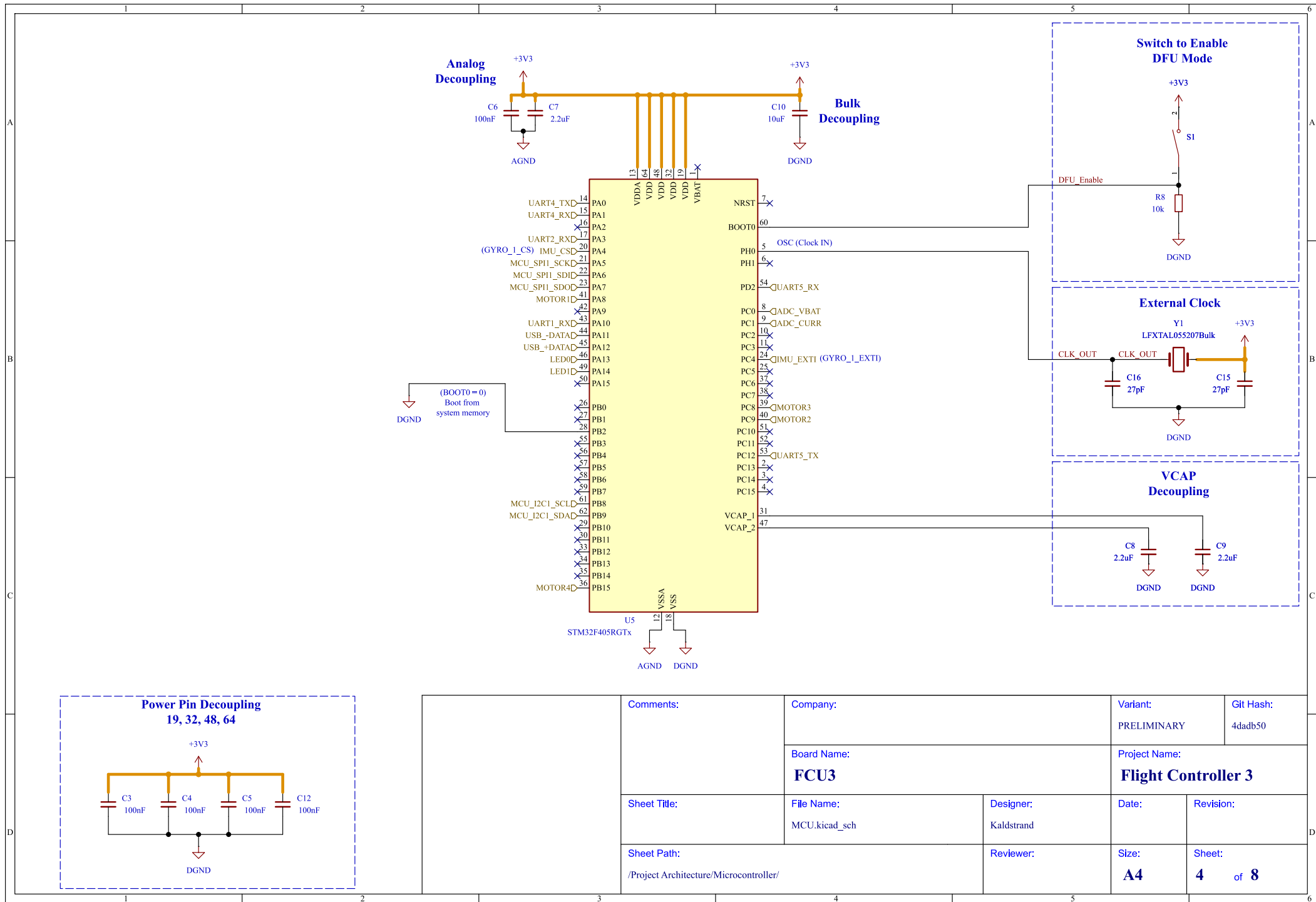
Grounding Strategy

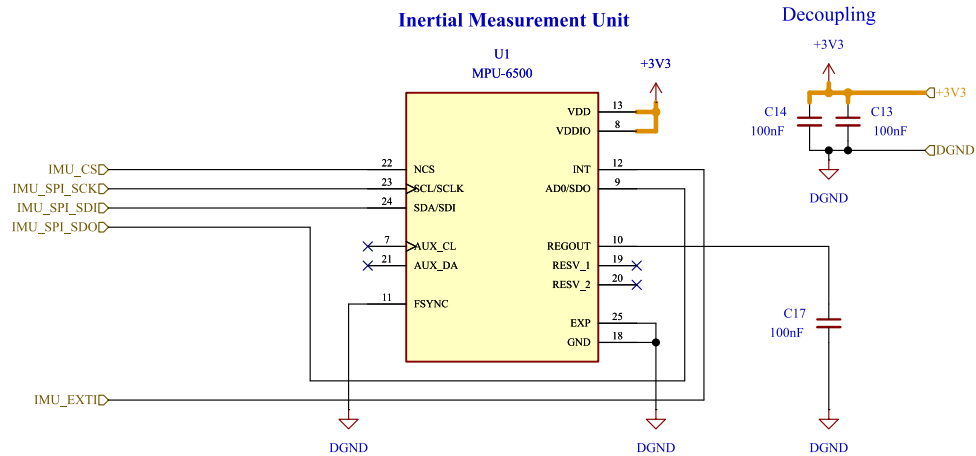
Ground Ties

Mounting Holes



Comments:	Company: Kaldstrand Electronics		Variant: PRELIMINARY	Git Hash: 4dad50
	Board Name: FCU3		Project Name: Flight Controller 3	
Sheet Title: Project Architecture	File Name: Project Architecture.kicad_sch	Designer: Kaldstrand	Date: 2025-01-12	Revision: + (Unreleased)
Sheet Path: /Project Architecture/		Reviewer:	Size: A3	Sheet: 3 of 8





	Comments:	Company:		Variant: PRELIMINARY	Git Hash: 4dadb50
		Board Name: FCU3		Project Name: Flight Controller 3	
	Sheet Title:	File Name: IMU.kicad_sch	Designer: Kaldstrand	Date:	Revision:
	Sheet Path: /Project Architecture/Inertial Measurement Unit/		Reviewer:	Size: A4	Sheet: 5 of 8

To-Do:

- Add all settings used in Vbat to 9V converter.
- Add all settings used in 3.3V LDO.

10-26V to 9V2A Step-Down Converter
File: stepdown_10-26vin_9v2a.kicad_sch

Vin

+9V

10-25.2V @ (0, 3, 100)A

V_USB

5V @ (0, 500, 3000)mA

D3
SS14-M3/5AT

D1
SS14

C1
1uF

JP4

LDO
3V3 @ (0, 50, 200)mA
U2
NCV4274AST33T3G

IN

OUT

GND

C20
10uF

+3V3

DGND

PGND

DESIGN NOTE:
JP4 is the Power Ground (PGND) to
Digital Ground (DGND) connection.

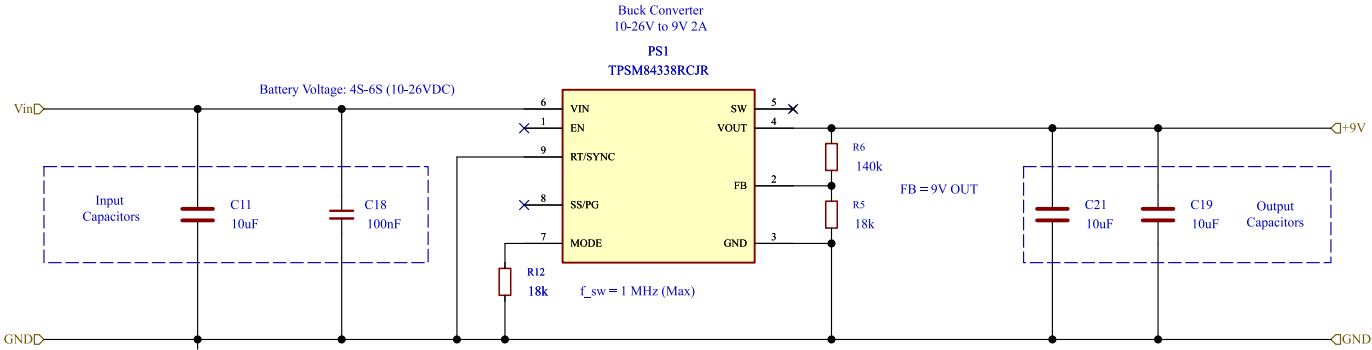
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		Board Name: FCU3		Project Name: Flight Controller 3	
	Sheet Title:	File Name: power.kicad_sch	Designer: Kaldstrand	Date:	Revision:
	Sheet Path: /Project Architecture/Power/	Reviewer:	Size: A4	Sheet: 6 of 8	

	Comments:	Company:		Variant:	Git Hash:
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		Board Name: FCU3		Project Name: Flight Controller 3	
	Sheet Title:	File Name: power.kicad_sch	Designer: Kaldstrand	Date:	Revision:
Sheet Path: /Project Architecture/Power/			Reviewer:	Size: A4	Sheet: 6 of 8

DESIGN NOTE:
 $R_{FBT} = (V_{OUT} - V_{REF}) / V_{REF} * R_{FBB}$
 $R_{FBT} = (9V - 0.6V) / 0.6V * 10k$
 $R_{FBT} = 140k$

 $V_{OUT} = 9V, V_{REF} = 0.6V, R11 = 10k, R6 = 140k$

To-Do:
Add design notes
explaining DC/DC settings.



	Comments:	Company:		Variant:	Git Hash:
				PRELIMINARY	4dad50
	Sheet Title:	Board Name:		Project Name:	
		FCU3		Flight Controller 3	
	Sheet Path:	File Name:	Designer:	Date:	Revision:
	/Project Architecture/Power/10-26V to 9V2A Step-Down Converter/	stepdown_10-26vin_9v2a.kicad_sch	Kaldstrand		
		Reviewer:	Size:	Sheet:	
			A4	8 of 8	

[9] Revision History

	Comments:	Company: Kaldstrand Electronics		Variant: PRELIMINARY	Git Hash: 4dad50
		Board Name: FCU3		Project Name: Flight Controller 3	
	Sheet Title: Revision History	File Name: Revision History.kicad_sch	Designer: Kaldstrand	Date: 2025-01-12	Revision: + (Unreleased)
	Sheet Path: /Revision History/		Reviewer:	Size: A4	Sheet: 9 of 8